SAYANTAN KHATUA

DATA ANALYST / BUSINESS ANALYST / BI ANALYST INTERN

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PROFESSIONAL SUMMARY

Business Analytics student who builds end-to-end analysis on real, messy datasets - from 8,000 marketing leads to 19.5 million taxi trips and $9.5 trillion in federal spending. Works validation-first in SQL and Python, tests findings statistically instead of reporting raw averages, and ships the result as Excel and Power BI dashboards on GitHub. Seeking a Data Analyst, Business Analyst or BI Analyst internship in India.

TECHNICAL SKILLS

SQL & Databases: MySQL, DuckDB - CTEs, window functions, joins, cohort and RFM queries

Python: Pandas, NumPy, Matplotlib, SciPy, Statsmodels - data cleaning, EDA, hypothesis testing, forecasting

BI & Excel: Power BI (star schema, DAX measures), Excel dashboards (PivotTables, charts), IBM Cognos Analytics

Statistics: Hypothesis testing (Wilcoxon, Mann-Whitney, chi-square), robust OLS regression, SARIMAX forecasting

Tools: Git / GitHub (iterative, documented commits), VS Code

PROJECTS

1. Federal Budget Execution Analysis github.com/sayantan-cyber/04-federal-budget-execution-analysis *Python, SQL, Excel*

- Analyzed U.S. federal budget execution across 888 agency-years (111 agencies, FY2017-2024) from real USAspending.gov data; built a validation-first Python pipeline (12 documented data-quality warnings) and 4 SQL analysis modules.
- Proved a year-end spending rush with a Wilcoxon signed-rank test ($p = 3.4e-14$, effect size $r = 0.53$) and built a holdout-validated forecast of agency obligation rates.
- Added a 58,282-row x 49-column bulk layer (FY2024, $9.519T reconciled to the agency series) showing the top 5 object classes absorb 76% of federal obligations; shipped an Excel dashboard with 3 native charts and 5 publication charts across 9 iterative commits.

2. NYC Taxi Demand Analytics github.com/sayantan-cyber/03-nyc-taxi-demand-analytics *Python, DuckDB, Statistics*

- Analyzed 19.5M NYC yellow-taxi trips ($546.8M in fares, Jan-Jun 2023) in DuckDB and Python; demand peaks Friday at 18:00, JFK is the busiest pickup zone (993,503 trips), and the top 10 of 265 zones hold 46.8% of revenue.
- Quantified tipping economics: card tips average 24.7% of fare and rush hour lifts tips by +1.1 percentage points ($p < 0.001$, robust OLS, $n = 300k$); beat the seasonal-naive baseline with a SARIMAX hourly forecast (MAPE 23.8% vs 28.4%).

3. Brazilian E-Commerce Revenue Analysis github.com/sayantan-cyber/01-ecommerce-revenue-analysis *SQL, Python, Power BI*

- Analyzed 99,441 real marketplace orders (R$13.22M revenue, Sep 2016-Oct 2018) with advanced SQL (CTEs, window functions, cohort retention, RFM segmentation) and Python EDA.
- Found late delivery (8.0% of orders) cuts review scores from 4.29 to 2.57, only 3.0% of customers ever reorder, and the top RFM segment (6.4% of customers) drives 14.3% of lifetime revenue.

4. Marketing Funnel & Lead Conversion github.com/sayantan-cyber/02-marketing-funnel-conversion *SQL, Python*

- Measured funnel conversion on 8,000 marketing-qualified leads: 10.5% became closed deals (842). Paid search (12.3%) and organic search (11.8%) outperformed social (5.6%) and email (3.0%); median sales cycle 14 days.
- Linked 45.1% of won deals to the e-commerce seller dataset via seller_id; flagged declared-revenue fields as unusable (94.7% zeros) rather than reporting them as insight.

EDUCATION & EXPERIENCE

Brainware University, West Bengal
BBA in Business Analytics - 2nd Year, 3rd Semester
Expected graduation: 2029
Scored 97.5% (19.5/20) in core analytical coursework class tests.

IBM Student Industry Experience Program
Selected for a two-day corporate visit to IBM Kolkata (September 2026) covering enterprise data workflows and digital transformation.
IBM Certifications (2026): Business Intelligence; Data Visualization with Python; Data Visualization with Cognos Dashboard Embedded.